

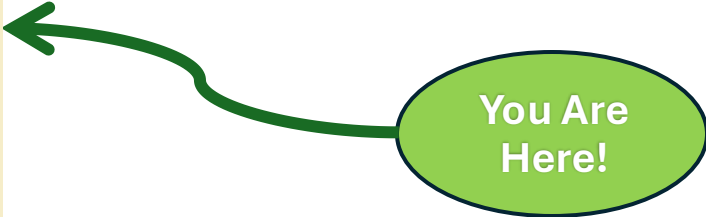
AI Fundamentals of Counterfactuals

Computational Foundations of Counterfactuals



Plan for the Day

Time	Topics
8:30 AM	Introduction
	<i>Hello and Introductions!</i>
8:35 AM	From Philosophy of Counterfactuals to XAI and User Studies
	<i>Philosophical & Psychological Fundamentals of Counterfactuals</i>
	<i>Getting Into the User Study Mindset</i>
	<i>Q&A</i>
9:10 AM	Computational Fundamentals of Counterfactuals in AI
	<i>Formalisation</i>
	<i>Current and Future Directions</i>
	<i>Q&A</i>
10:00 AM	COFFEE (10:00-10:30)
10:30 AM	How to Run a User Study
	<i>A Roadmap: From Motivation to Results</i>
	<i>Q&A</i>
11:15 AM	Hands-on
	Hands-on: <i>Computing Counterfactuals</i>
	Hands-on: <i>Analyzing a User Study</i>
12:20 PM	Closing Session, Discussion and Final Q&A
TUTORIAL END	



You Are Here!

Detailed Outline



Formalization



Modeling Approaches



Key Constraints

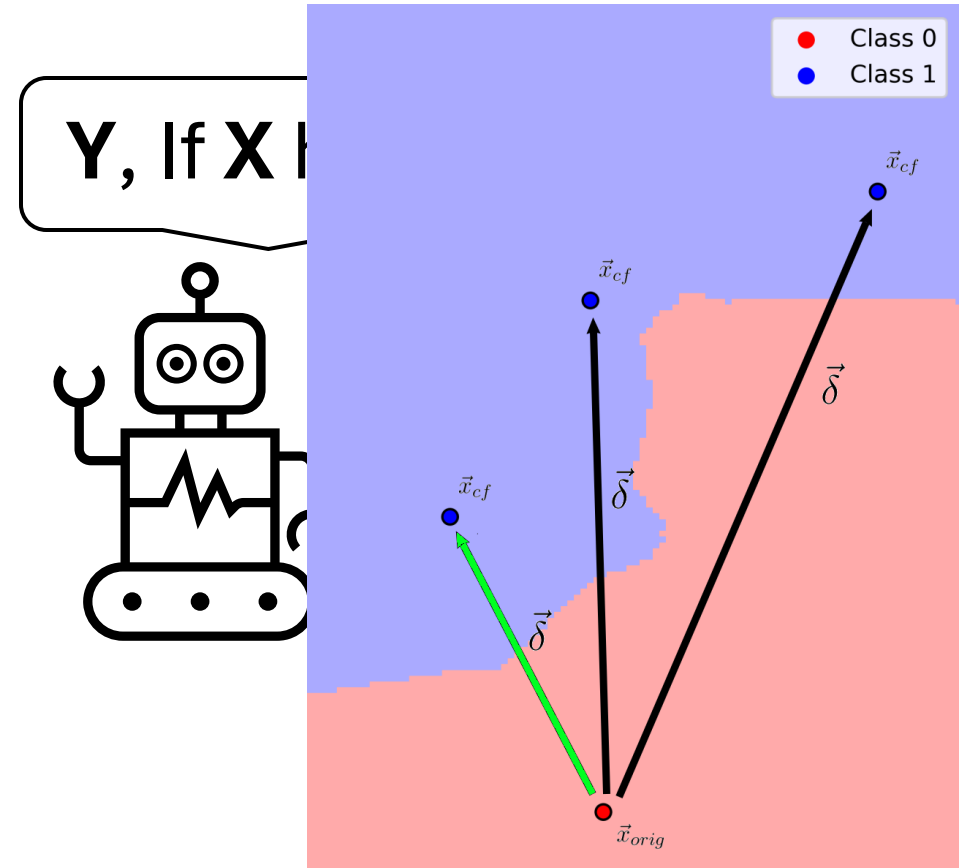


Hands-On: Counterfactuals in Python

Formalization of Counterfactuals

- Two properties:

1. Contrastive
2. Proximity/Closeness/Cost
 - Often a p-norm – but is this realistic?
=> Domain specific!



Modeling Approaches

- Optimization problem [\[Wachter 2017\]](#):

$$\arg \min_{\vec{x}_{cf} \in \mathbb{R}^d} \boxed{\ell(h(\vec{x}_{cf}), y_{cf})} + C \cdot \boxed{\theta(\vec{x}_{cf}, \vec{x}_{orig})}$$

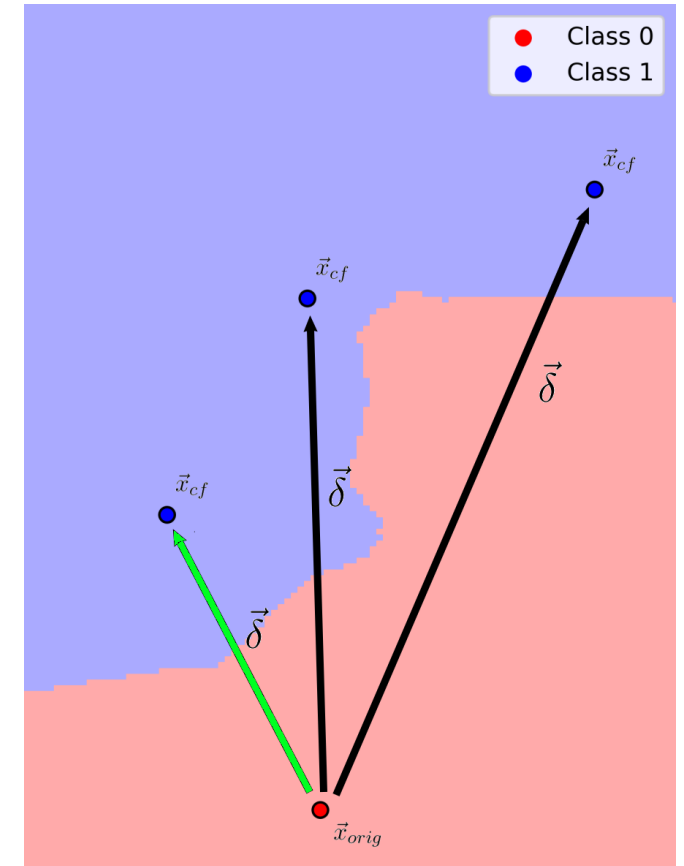
↑**Contrastive** ↑**Cost/Proximity**

$$\text{Change } \vec{\delta} = \vec{x}_{cf} - \vec{x}_{orig}$$

In constraint form:

$$\arg \min_{\vec{x}_{cf} \in \mathbb{R}^d} \boxed{\theta(\vec{x}_{cf}, \vec{x}_{orig})} \quad \text{s.t.} \quad \boxed{h(\vec{x}_{cf}) = y_{cf}}$$

↑**Cost/Proximity** ↑**Contrastive**

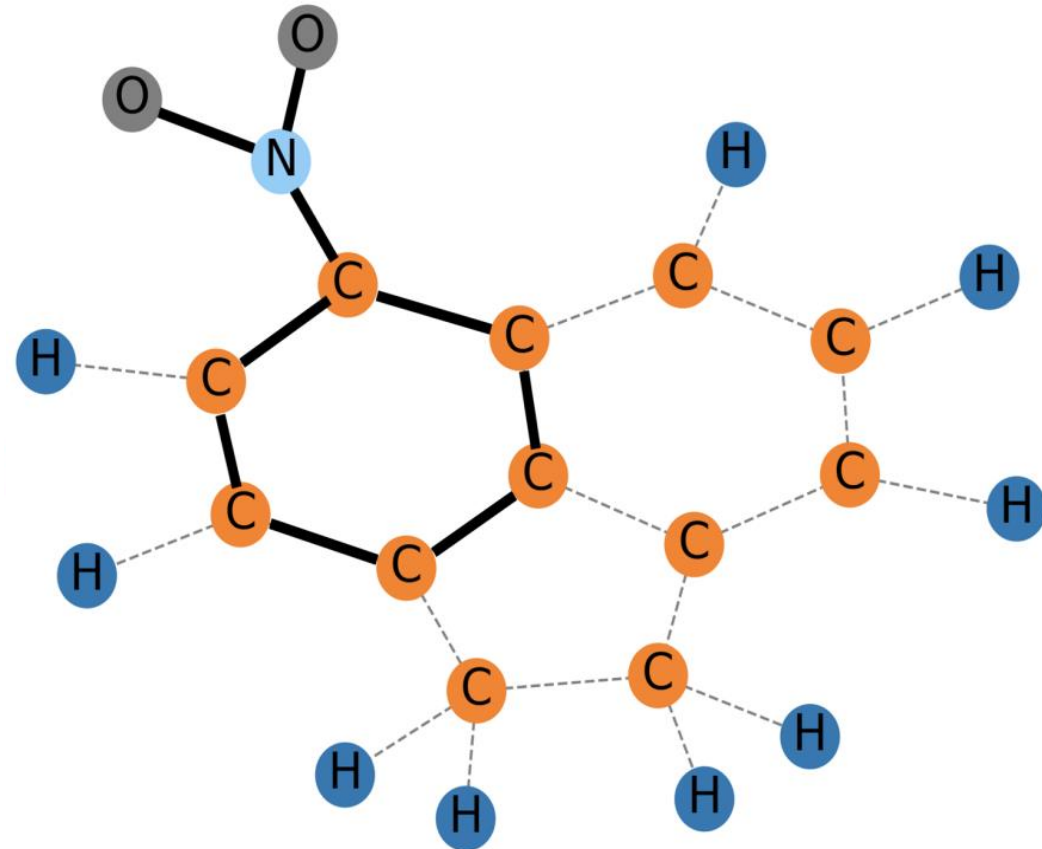


Modeling Approaches

- How to solve the optimization problems?

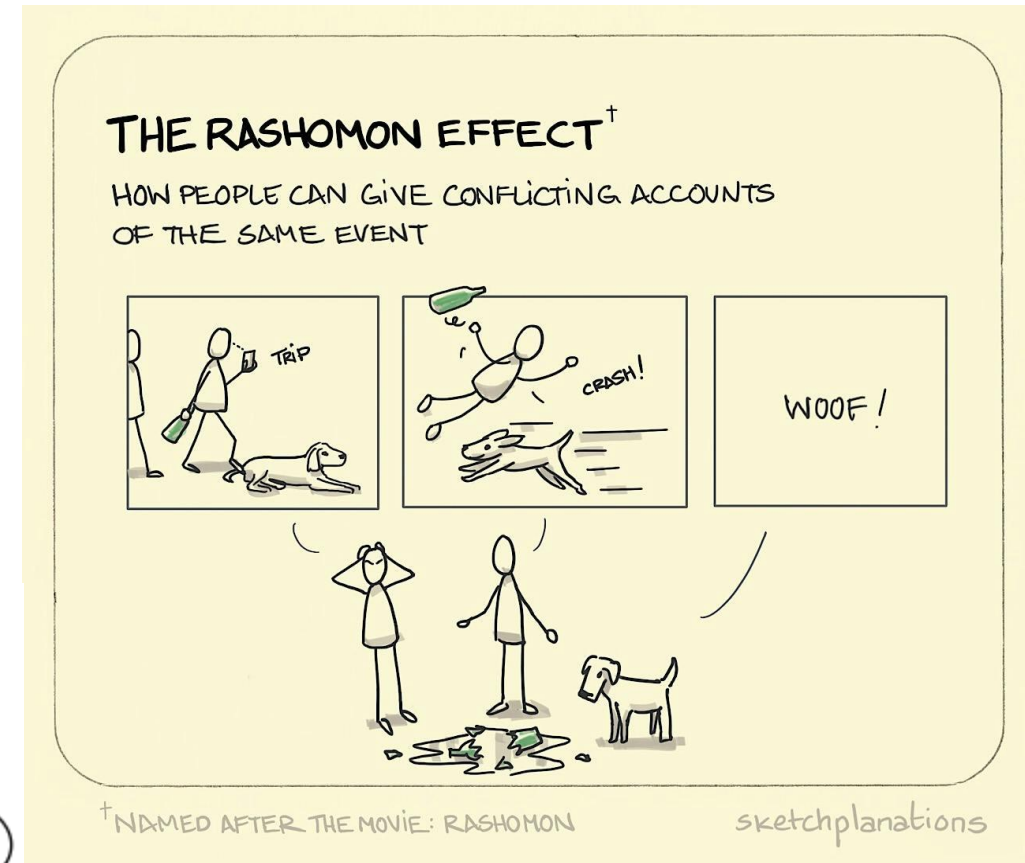
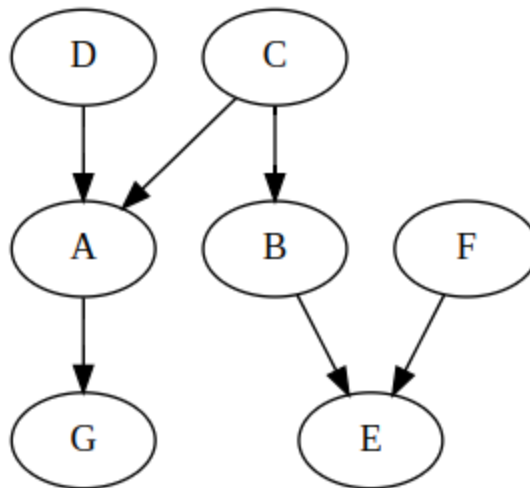
$$\arg \min_{\vec{x}_{\text{cf}} \in \mathbb{R}^d} \ell(h(\vec{x}_{\text{cf}}), y_{\text{cf}}) + C \cdot \theta(\vec{x}_{\text{cf}}, \vec{x}_{\text{orig}})$$

- Black-box solvers
- Gradient descent
- Domain-specific methods (images, text, graphs, etc.)



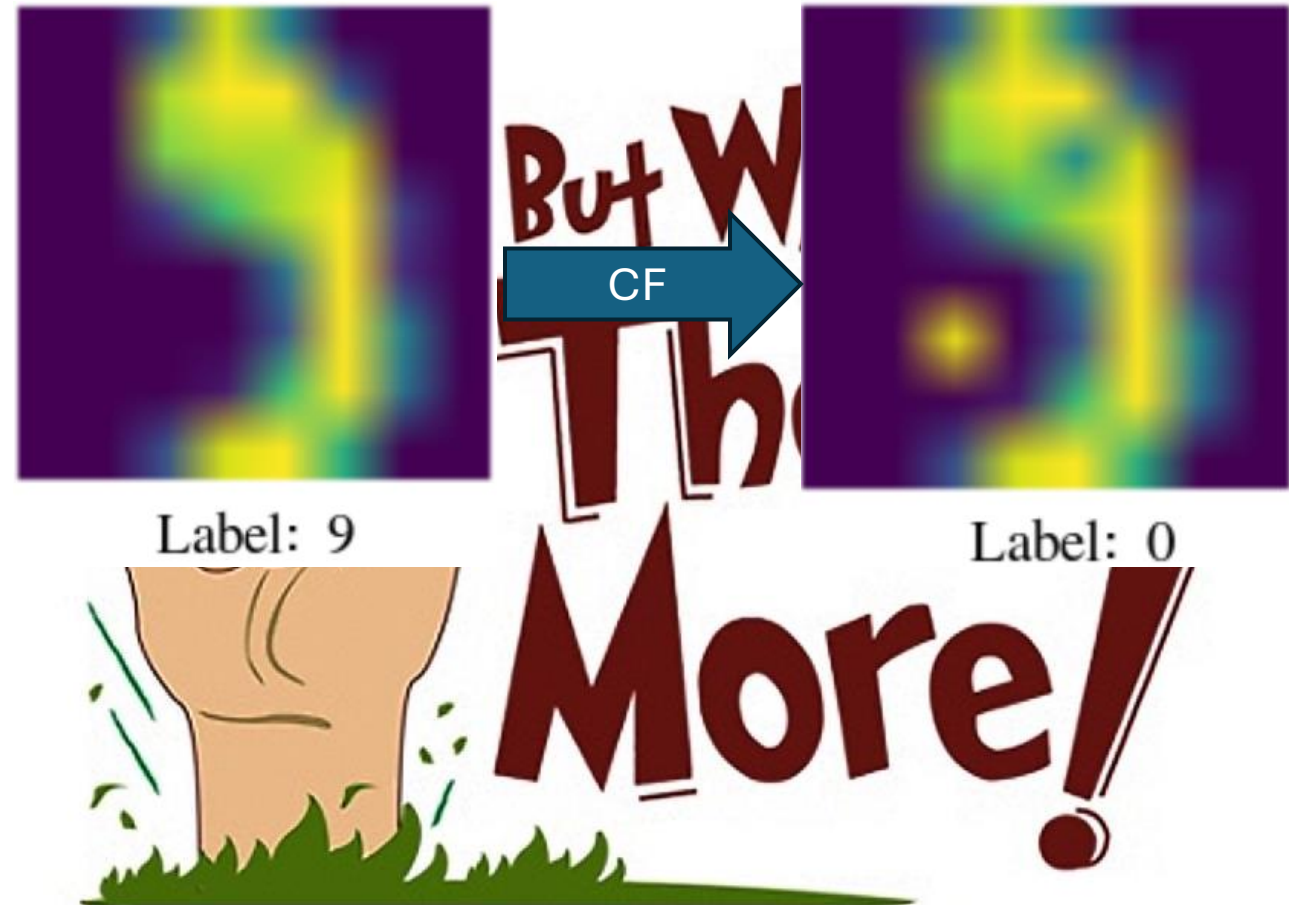
Things to keep in Mind

- Uniqueness?
=> "Rashomon Effect"
- What about Causality? [\[Karimi 2021\]](#)
=> Often, feature independence is assumed!
Incorporate Causal models!



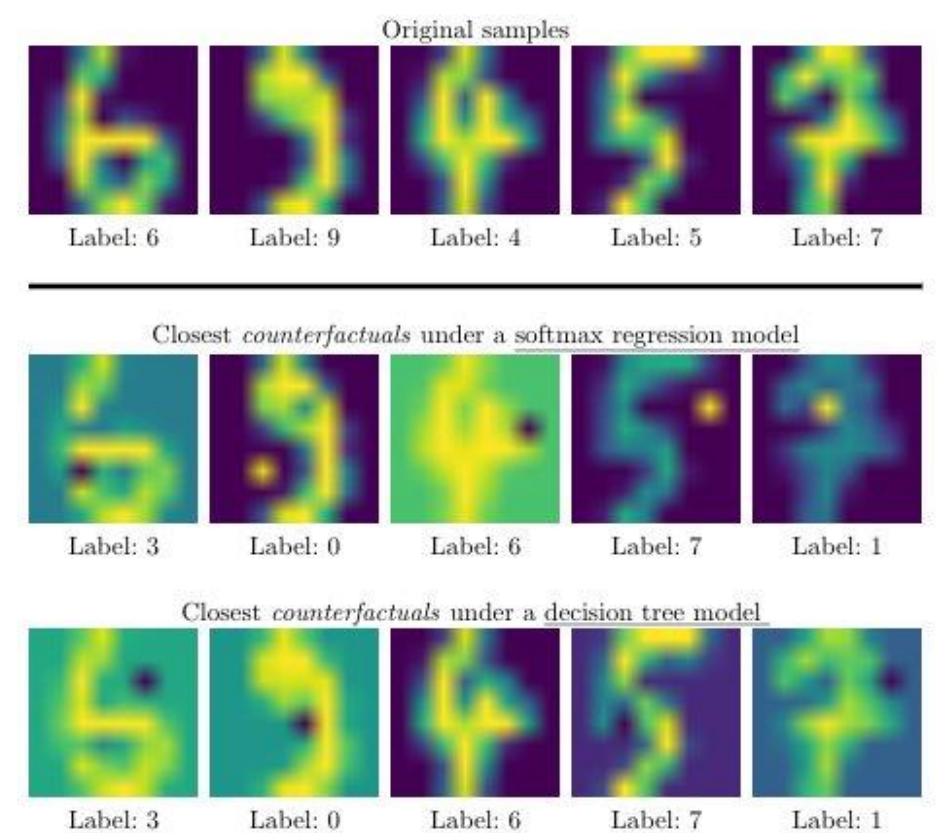
Fundamentals & Key Constraints

- Contrastivity and Cost/Proximity are not enough!
 - e.g. CF $\neq$ Adversarial
- Other important aspects:
 - Plausibility
 - Actionability
 - Diversity
 - ...



Plausibility & Actionability

- Problem: Classic CFs [\[Wachter 2017\]](#) often adversarialials
=> Not useful in practice! [\[Smyth 2022\]](#)
- **CFs must be plausible and actionable**



Plausibility & Actionability

- Two popular & general approaches:

- FACE [\[Poyiadzi 2020\]](#)
- Density constraints [\[Artelt 2020\]](#)

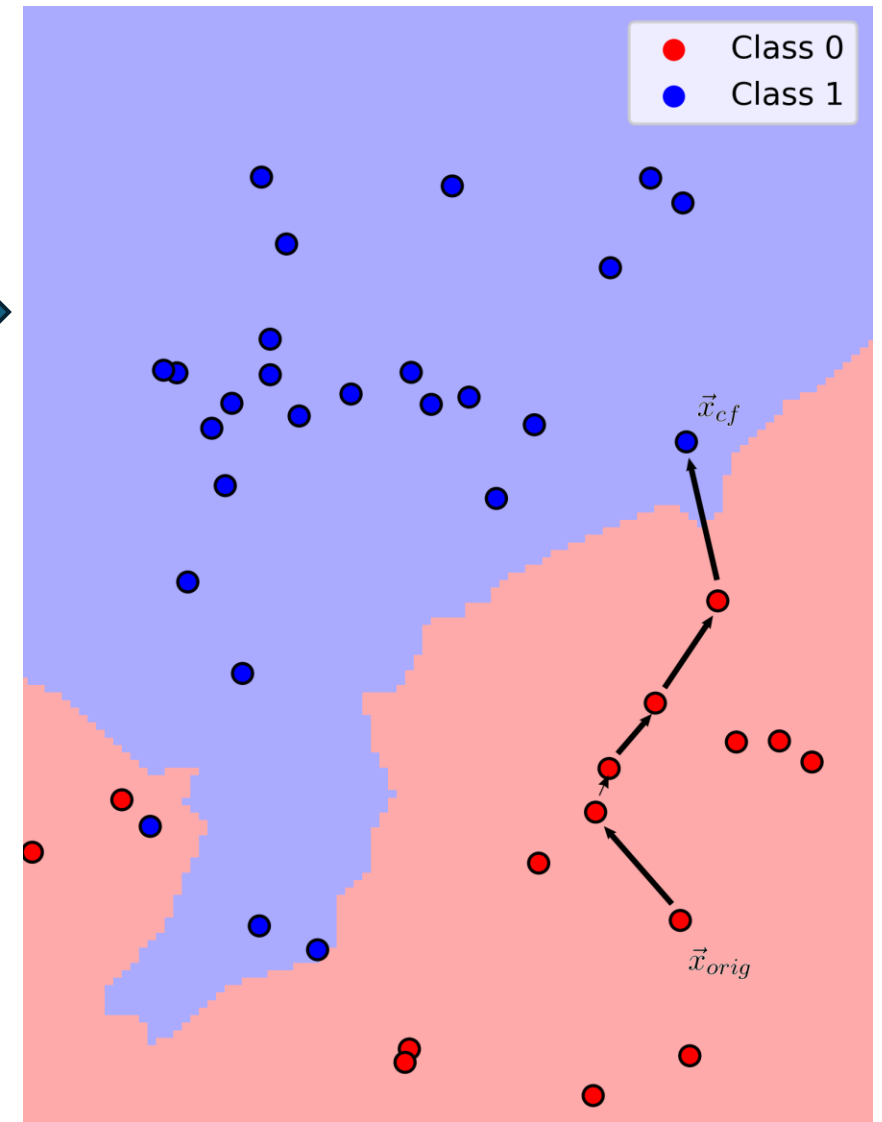
$$\arg \min_{\vec{x}_{cf} \in \mathbb{R}^d} \theta(\vec{x}_{cf}, \vec{x}_{orig})$$

$$\text{s.t. } h(\vec{x}_{cf}) = y_{cf}$$

$$\hat{p}_{y_{cf}}(\vec{x}_{cf}) \geq$$

δ

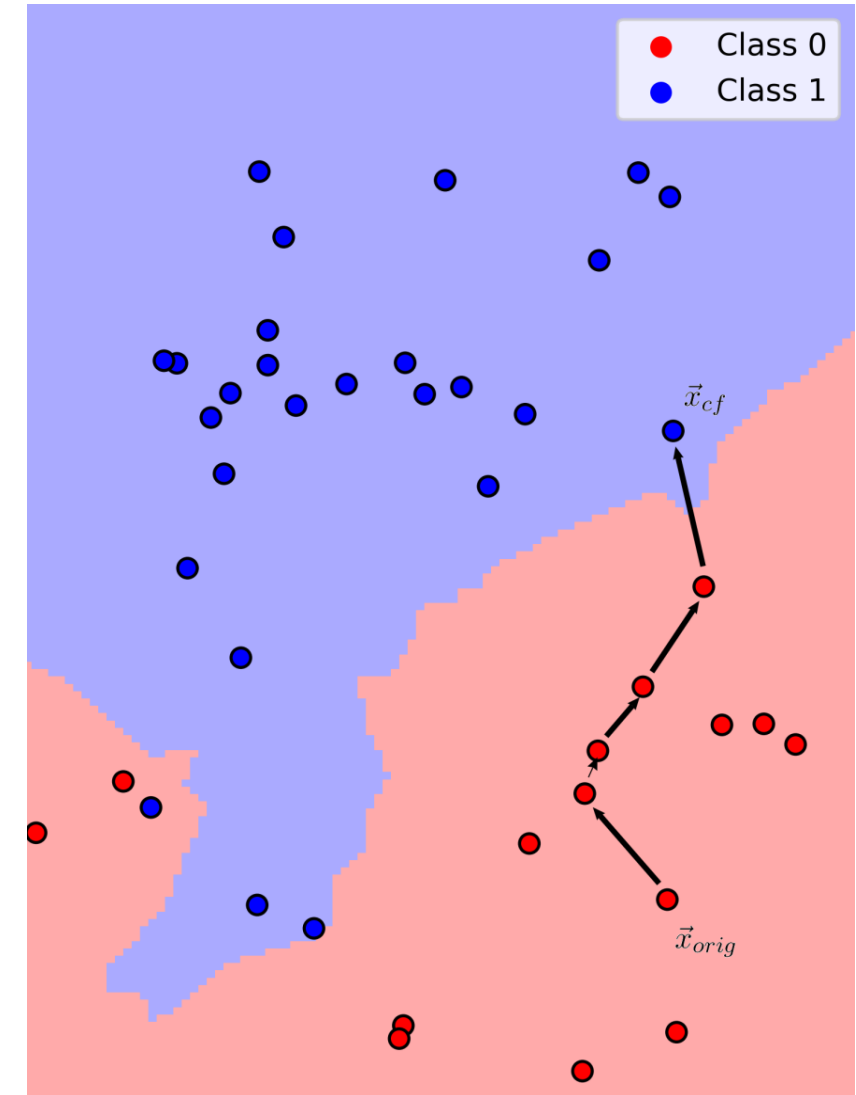
lower-bound on density



FACE: Feasible, Actionable CFs

[Poyiadzi 2020]

- Sequence of changes
- Limit feasible set to observed data
 - "No feature independence"
 - Actionable & Plausible
- Graph: KDE, k-NN, or ϵ -graph
=> Find shortest path

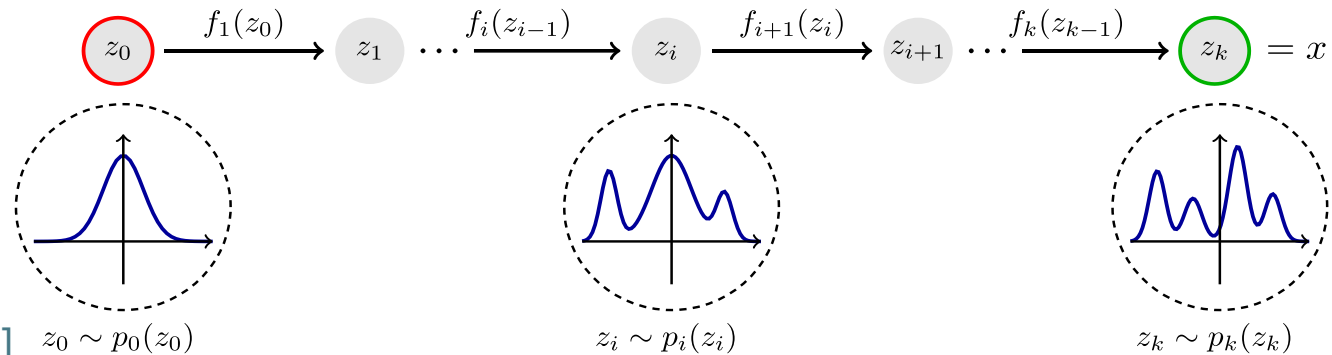


Density Constraints

- **CF in high-density region**

- Methods:

- GMMs [\[Artelt 2020\]](#)
- KDE [\[Förster 2021\]](#)
- GANs [\[Mertes 2022\]](#)
- Normalizing Flows [\[Wielopolski 2024\]](#)



$$\arg \min_{\vec{x}_{\text{cf}} \in \mathbb{R}^d} \theta(\vec{x}_{\text{cf}}, \vec{x}_{\text{orig}})$$

$$\text{s.t. } h(\vec{x}_{\text{cf}}) = y_{\text{cf}}$$

$$\hat{p}_{y_{\text{cf}}}(\vec{x}_{\text{cf}}) \geq$$

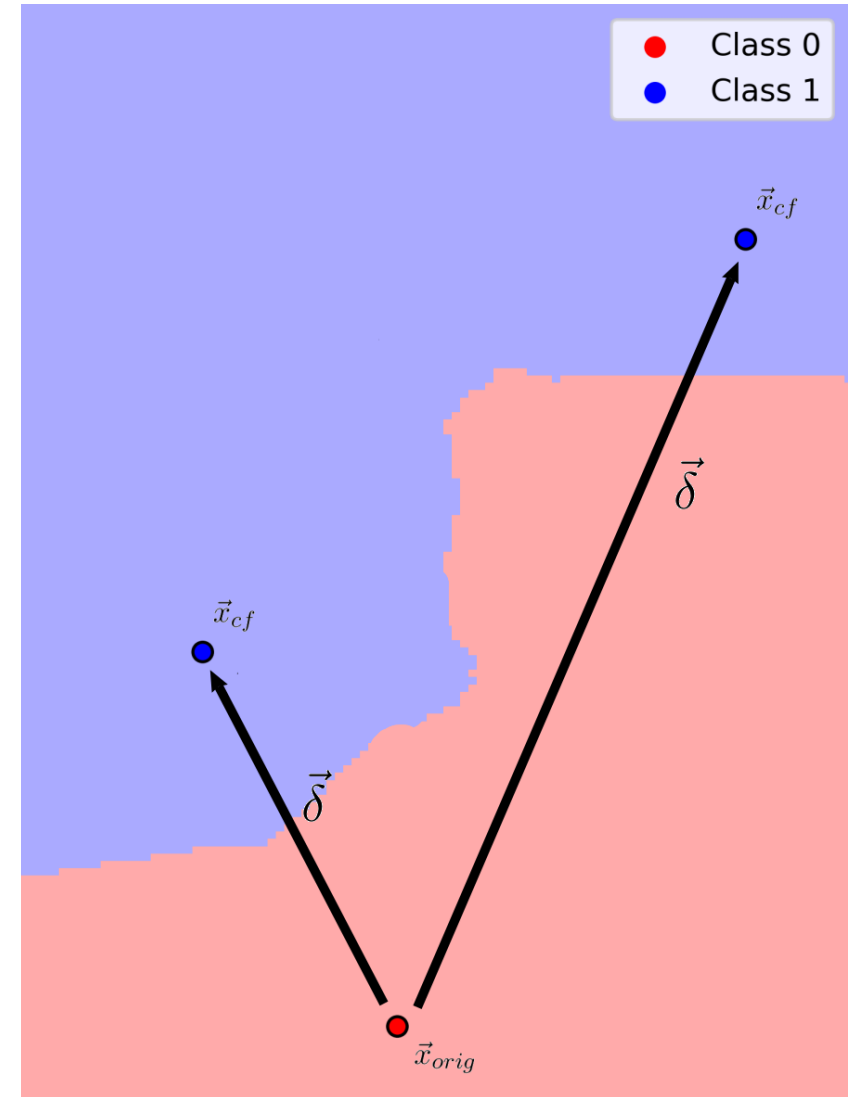
$$\underbrace{\delta}_{\text{lower-bound on density}}$$

lower-bound on density

Diversity

- Missing uniqueness
=> "Rashomon Effect"
- Generate multiple "different" CFs
 - Let the user choose!
 - More robust? [\[Leofante 2024\]](#)

RQ: How to model diversity?



Diversity

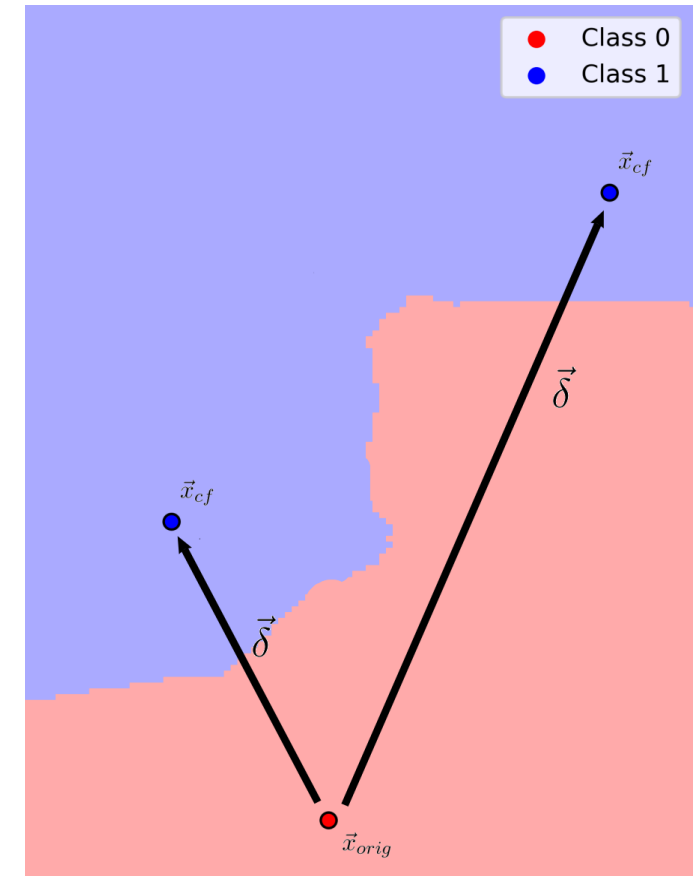
- Yet another constraint:
 - Quantify difference between CFs (e.g. cost, common features, etc.)

- Toolbox: DiCE [\[Mothilal 2020\]](#)
=> See Hands-on session

$$\arg \min_{\{\vec{x}_{cf_i} \in \mathbb{R}^d\}} \underbrace{\sum_i \ell(h(\vec{x}_{cf_i}), y_{cf})}_{\text{Constrasting}} + \underbrace{C_1 \sum_i \theta(\vec{x}_{cf_i}, \vec{x}_{orig})}_{\text{Complexity}} + \underbrace{C_2 \sum_{i,j} \psi(\vec{x}_{cf_i}, \vec{x}_{cf_j})}_{\text{Diversity}}$$

Summary & Conclusion

- Two important properties:
 - Contrastive
 - Cost/Proximity
- Key constraints:
 - Plausibility & Actionability
 - Diversity



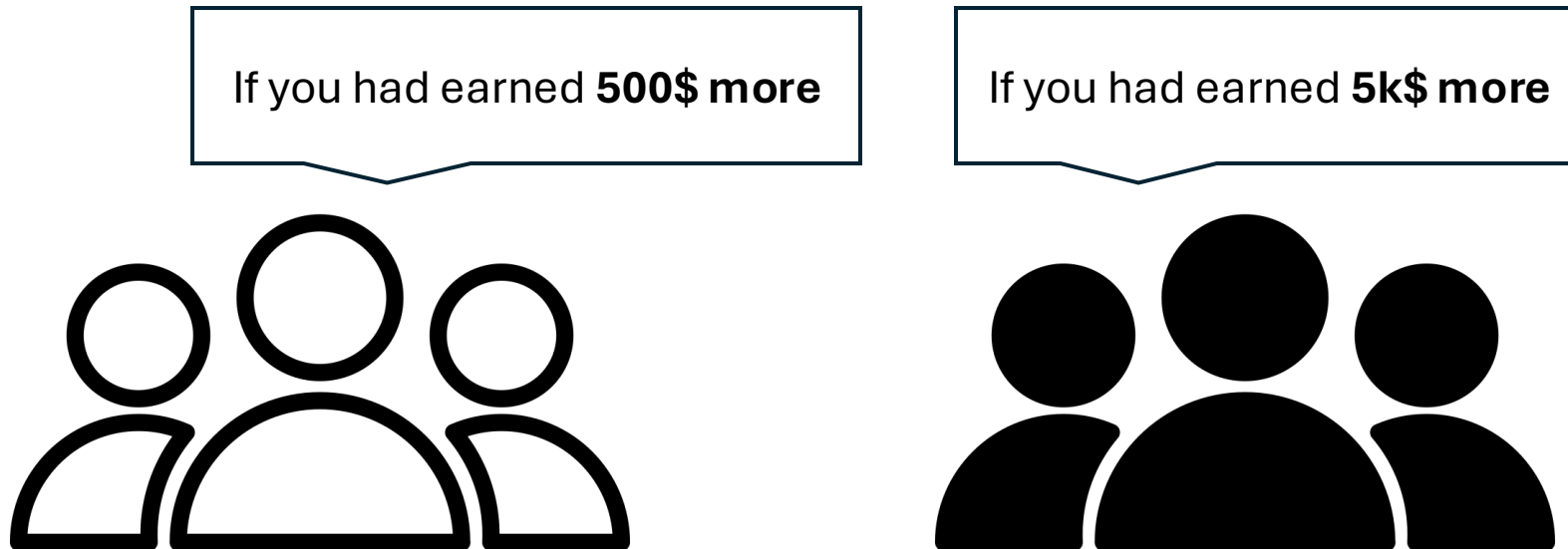
Other & Current Topics

1. Fairness
2. Robustness
3. (Adversarial) Manipulations
4. Group Counterfactuals



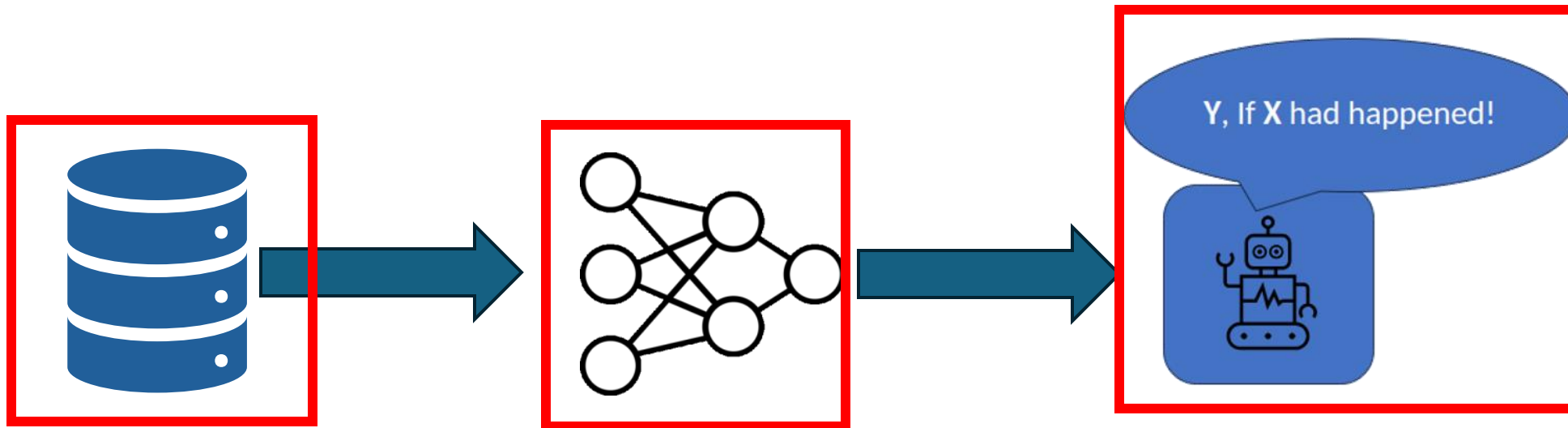
Fairness

- How can a CF be unfair?
=> Differences in the cost of recourse (global vs. local)
- Group vs. Individual fairness [[Slack 2021](#), [Sharma 2020/21](#), [Kügelgen 2022](#), [Artelt 2023](#)]



Approaches to Fix Fairness Issues

- Fix model [[Sharma 2020/21](#)]
- Fix CF generator [[Kügelgen 2022](#), [Artelt 2023](#)]
- Fix data [[Artelt 2024](#)]



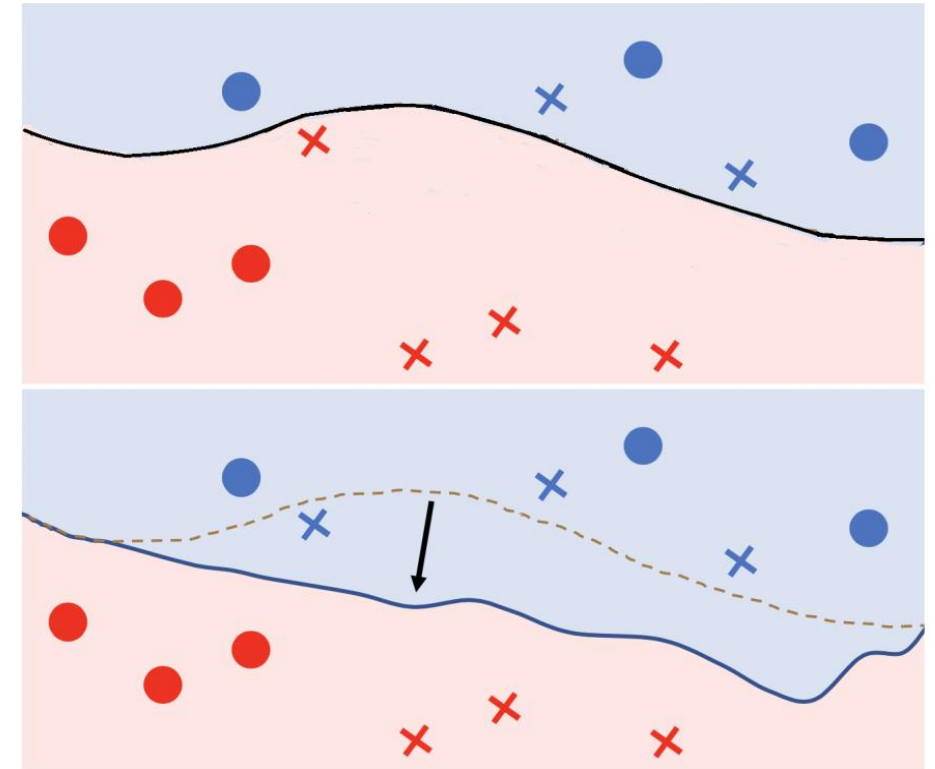
Make the Model Fair [\[Sharma 2020/21\]](#)

- Define cost of recourse for a group "g" as "Burden":

$$Burden(g) = \mathbb{E}_g[d(\mathbf{x}, \mathbf{c}^*)]$$

- Add fairness loss to training loss:

$$\mathcal{L}_{fairness} = \mathbb{E}_{\mathbf{x} | s(\mathbf{x})=a} [d(\mathbf{x}, \mathcal{B})] - \mathbb{E}_{\mathbf{x} | s(\mathbf{x})=b} [d(\mathbf{x}, \mathcal{B})]$$



Open Questions

- Which approach to use?
=> Limitations?
- Formal guarantees?
=> Impossibility results?
- ...

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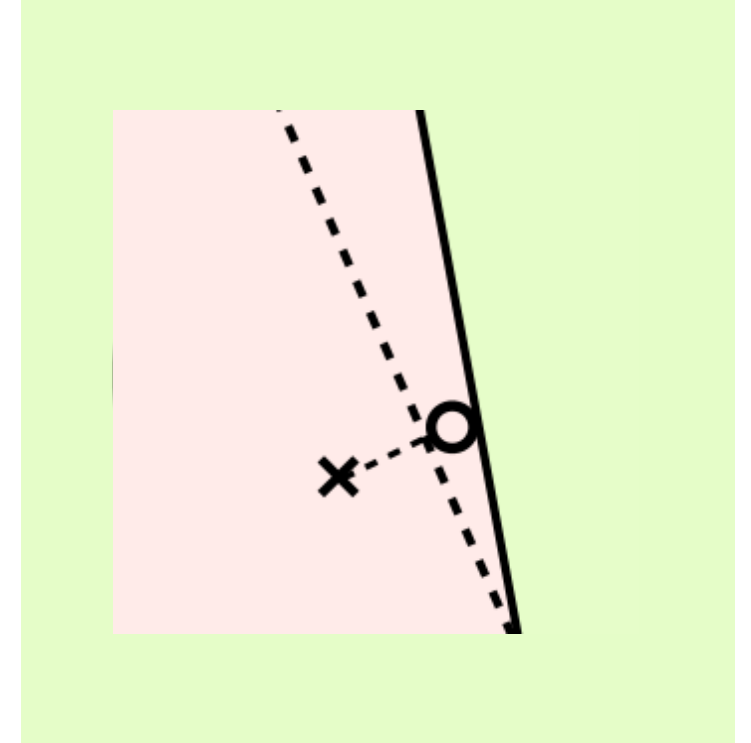


Robustness

- **(Missing) robustness** of CFs [[Jiang 2024](#), [Ferrario 2022](#), [Artelt 2021](#)]
 - Change in explanation
 - Change in cost
- *Input perturbations* (see individual fairness)
- *Model change* (over time)

=> Bad for the user!

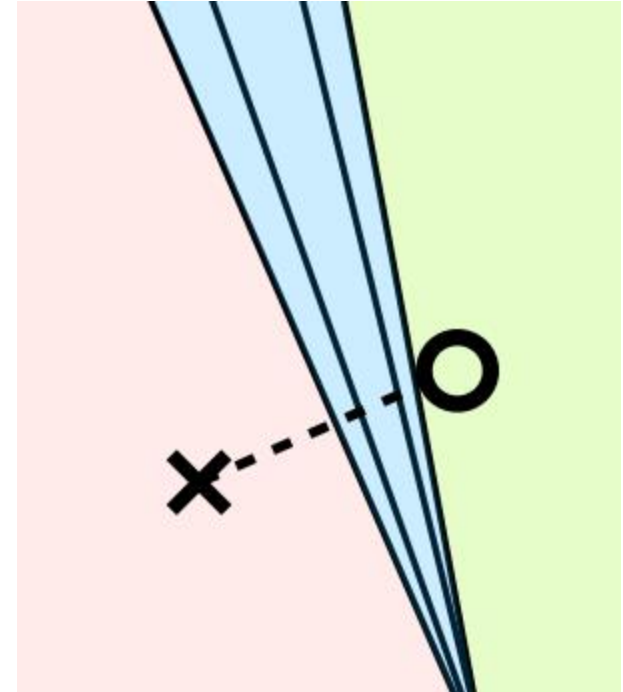
RQ: How to achieve robustness? 🤔



[Leofante 2023]

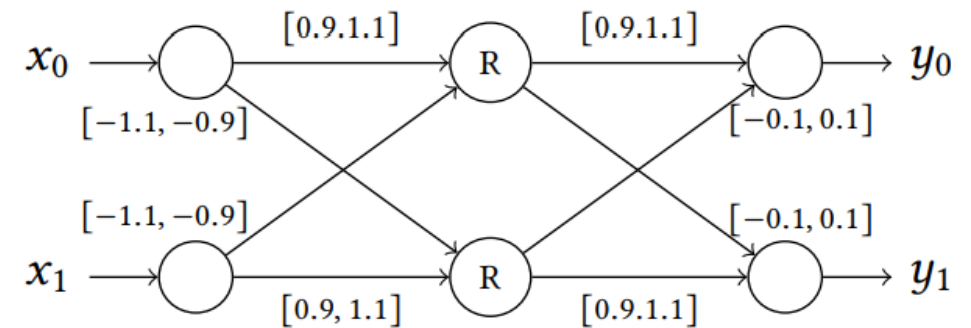
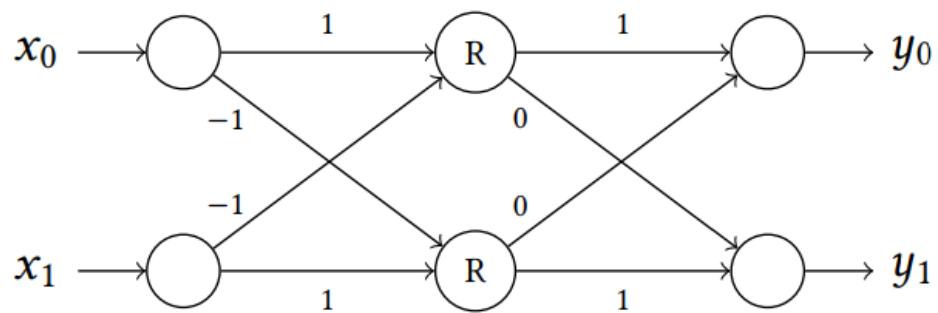
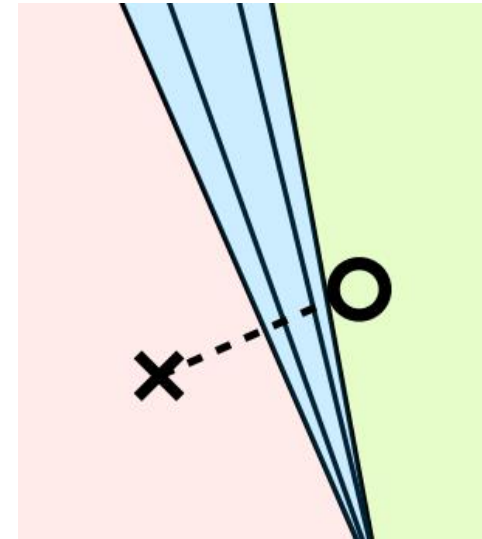
RQ: How to make CFs Robustness? 🤔

- Input changes:
 - Plausibility [[Artelt 2020-21](#), [Zhang 2023](#)]
 - Min-Max Objective [[Dominguez-Olmedo 2022](#)]
 - Diversity [[Leofante 2024](#)]
- Model changes
 - Plausibility [[Pawelczyk 2020](#)]
 - Possible model changes -> MILPs [[Jiang 2023-24](#)]



Robustness wrt. Model Changes [\[Jiang 2023-24\]](#)

- Consider ranges of parameters
- MILPs for computing & certifying robustness of CFs



Open Questions

- Cost of robustness
 - How much to sacrifice?
- Robustness and other aspects (e.g. plausibility, fairness, etc.)
 - Impossibility statements?
-

=> Check out Survey on Robustness [\[Jiang 2024\]](#)

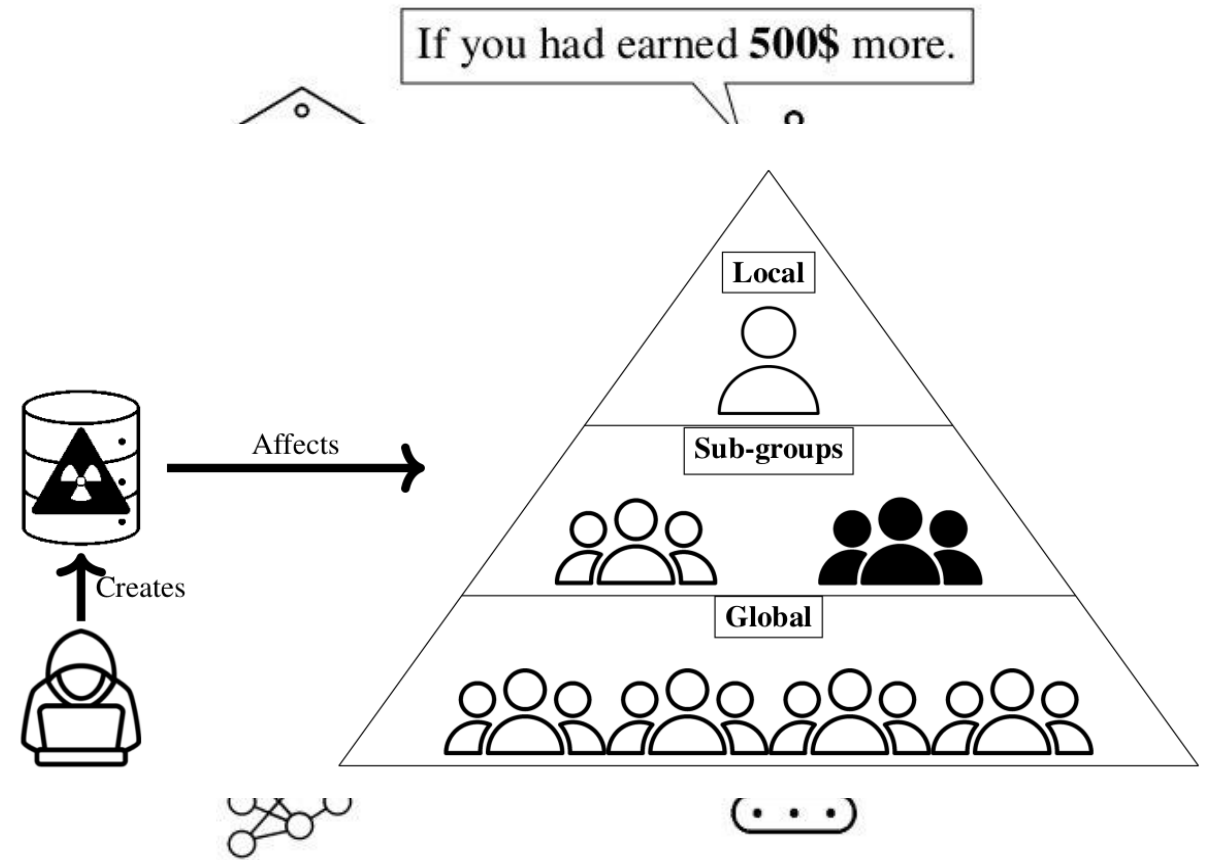
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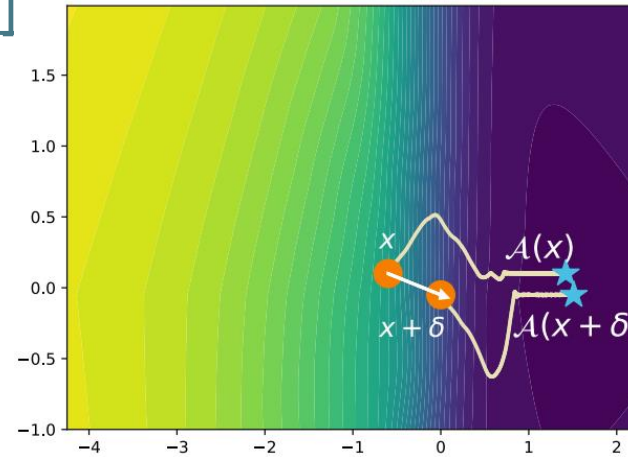
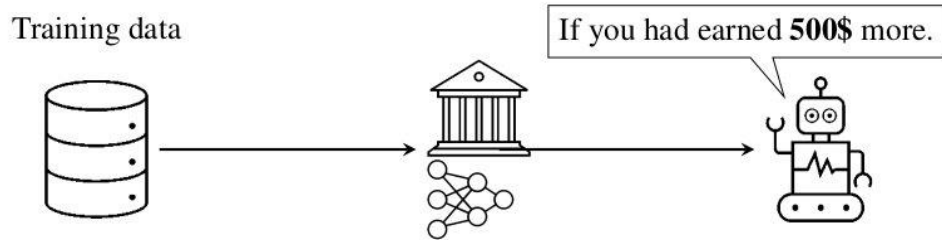
(Adversarial) Manipulations

- Can CFs be manipulated?
 - Still trustworthy?
- Attack goals:
 - Cost of recourse
 - Fairness
 - Robustness
 -
- Global vs. local attack

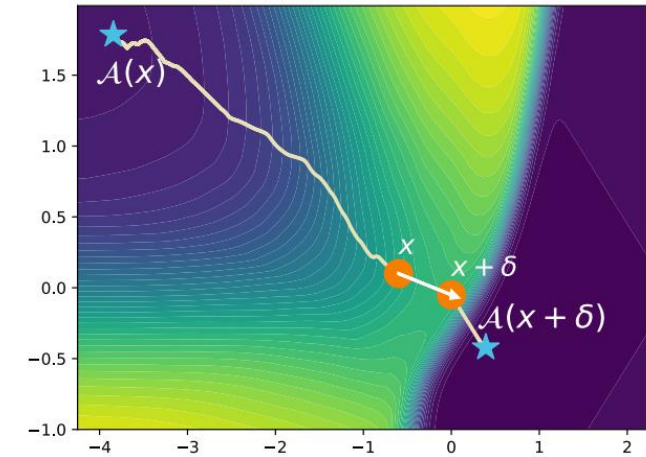


Manipulating CFs

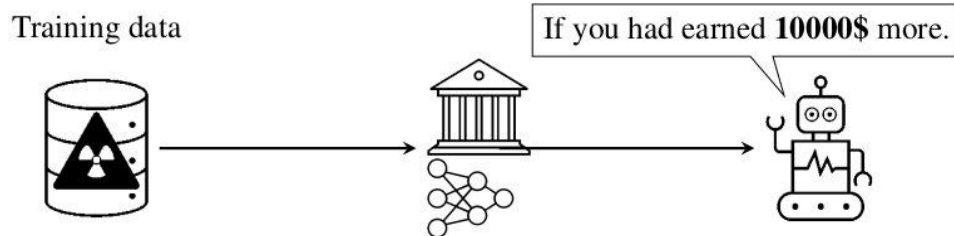
- Adversarial training [\[Slack 2021\]](#)
 - Introduce "unfairness"
- Data poisoning [\[Artelt 2024\]](#)
 - Increase cost of recourse



(a) Training with BCE Objective

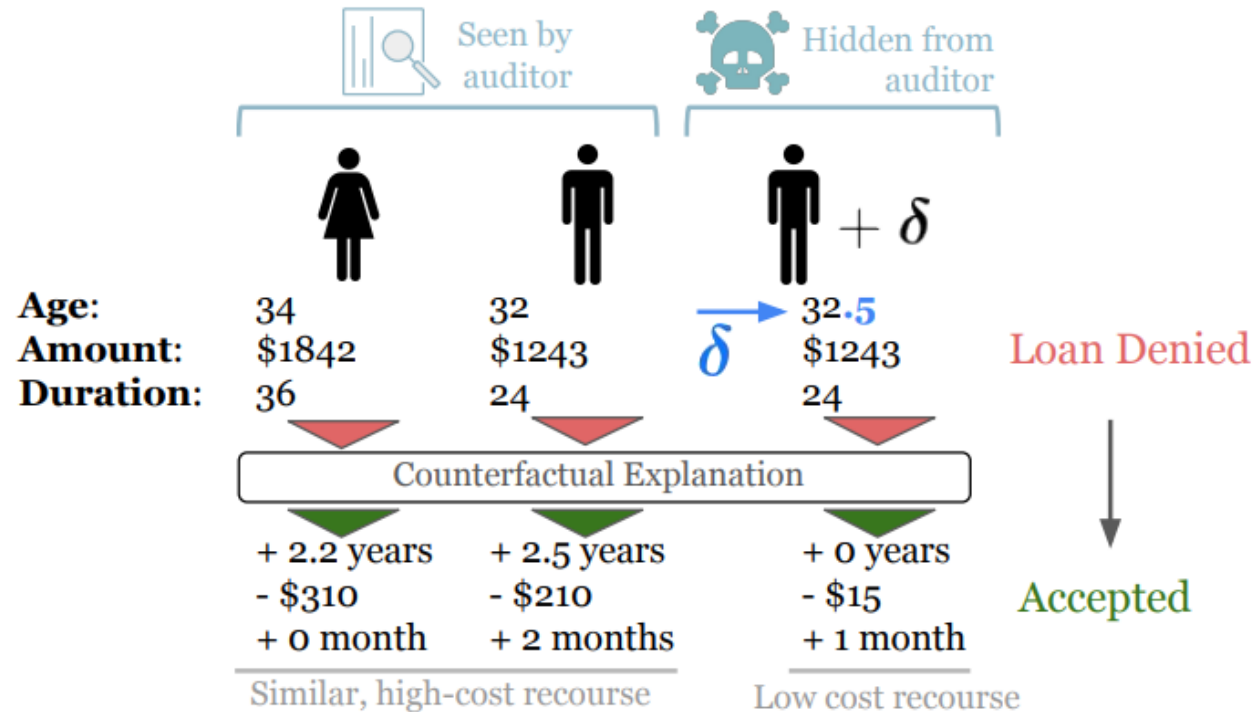


(b) Training Adversarial Model



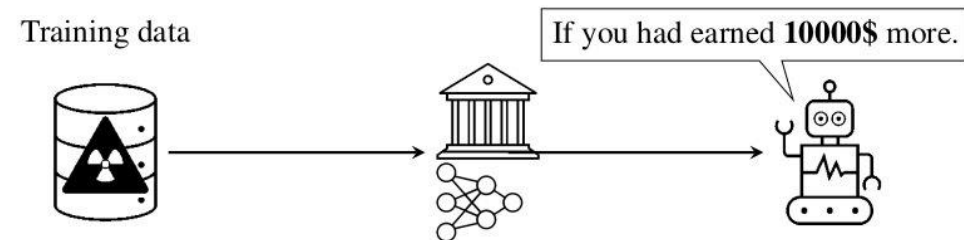
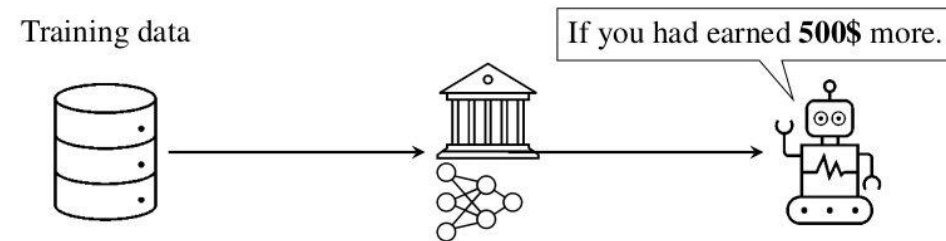
Adversarial Training [\[Slack 2021\]](#)

- **"Backdoor"**: Some small perturbation leads to lower cost recourse



Data Poisoning [\[Artelt 2024\]](#)

- Add poisonous samples to training data
=> Increase cost of recourse



Defense against Manipulations

- Observation: Additional density constraints help [\[Artelt 2024\]](#)
- **Open questions!** 

Other & Current Topics

- Semi-Factuals
- Fairness
- Robustness
- (Adversarial) Manipulations
- Group Counterfactuals



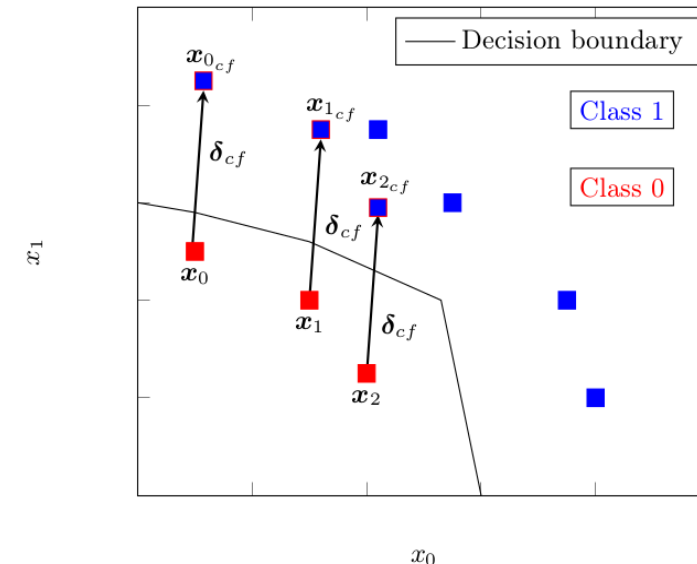
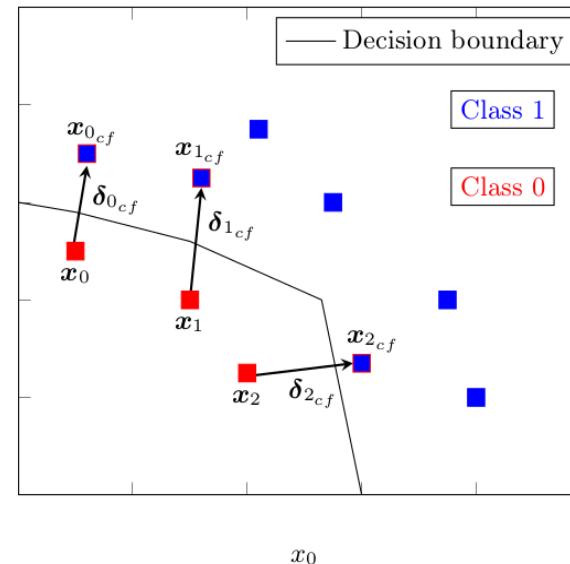
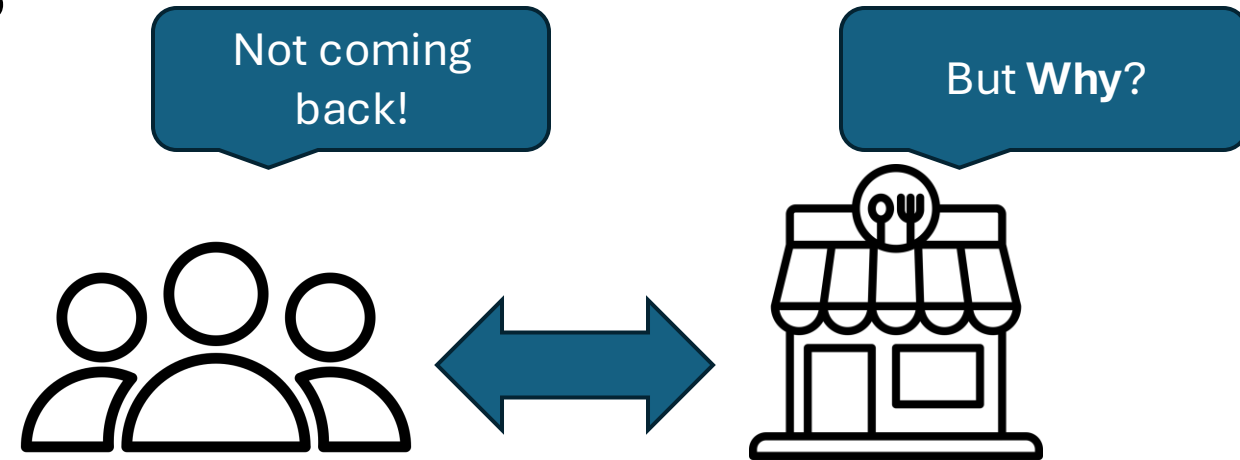
Group Counterfactuals

- One CF for many instances
- Real-world problems, e.g. Customer repurchase [\[Artelt 2024\]](#)

New challenges:

- Clustering/Grouping of instances

Current approaches [\[Kanamori 2022, Ley 2023, Warren 2023, Artelt 2023/24\]](#)



Group Counterfactuals

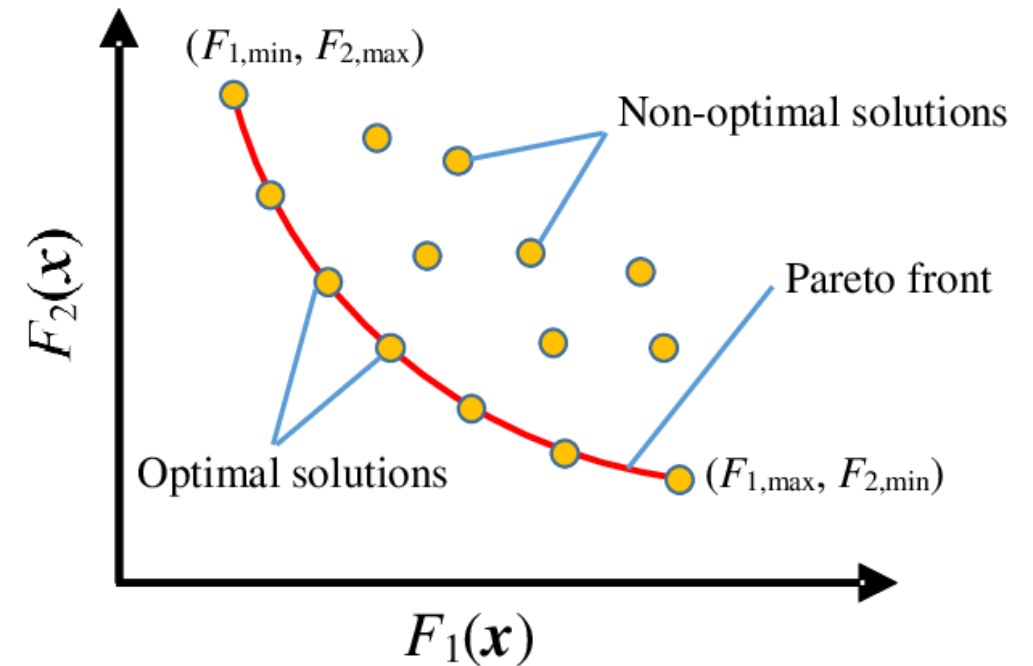
- Formalization [\[Artelt 2024\]](#):

$$\min_{\vec{\delta}_{\text{cf}} \in \mathbb{R}^d} \left(\underbrace{\theta(\vec{\delta}_{\text{cf}})}_{\text{Cost/Proximity}}, \underbrace{\sum_{\vec{x}_i \in \mathcal{D}} \ell(h(\vec{x}_i \oplus \vec{\delta}_{\text{cf}}), y_{\text{cf}})}_{\text{Contrastive}} \right)$$

- Feasibility?

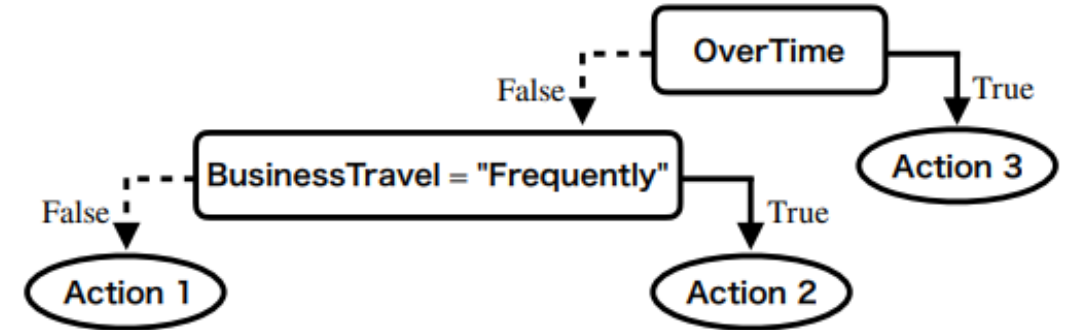
=> Pareto-optimal solutions!

What about other constraints? 🤔



Group Counterfactuals

- Grouping given?
- Grouping and CFs
 - All-in-One [\[Kanamori 2022\]](#)
 - Separate [\[Artelt 2024\]](#)



	HowToChange	Effectiveness	
		Cost	Flip rate
Action 1	MonthlyIncome : + 1282\$	0.17	83 %
Action 2	BusinessTravel : "Frequently" → "Rarely"	0.19	80 %
Action 3	OverTime : True → False	0.27	86 %

(a) Cluster with $d(\vec{\delta}_{cf_i}, \vec{\delta}_{cf_j}) = \frac{\vec{\delta}_{cf_i}^\top \vec{\delta}_{cf_j}}{\|\vec{\delta}_{cf_i}\|_2 \|\vec{\delta}_{cf_j}\|_2}$

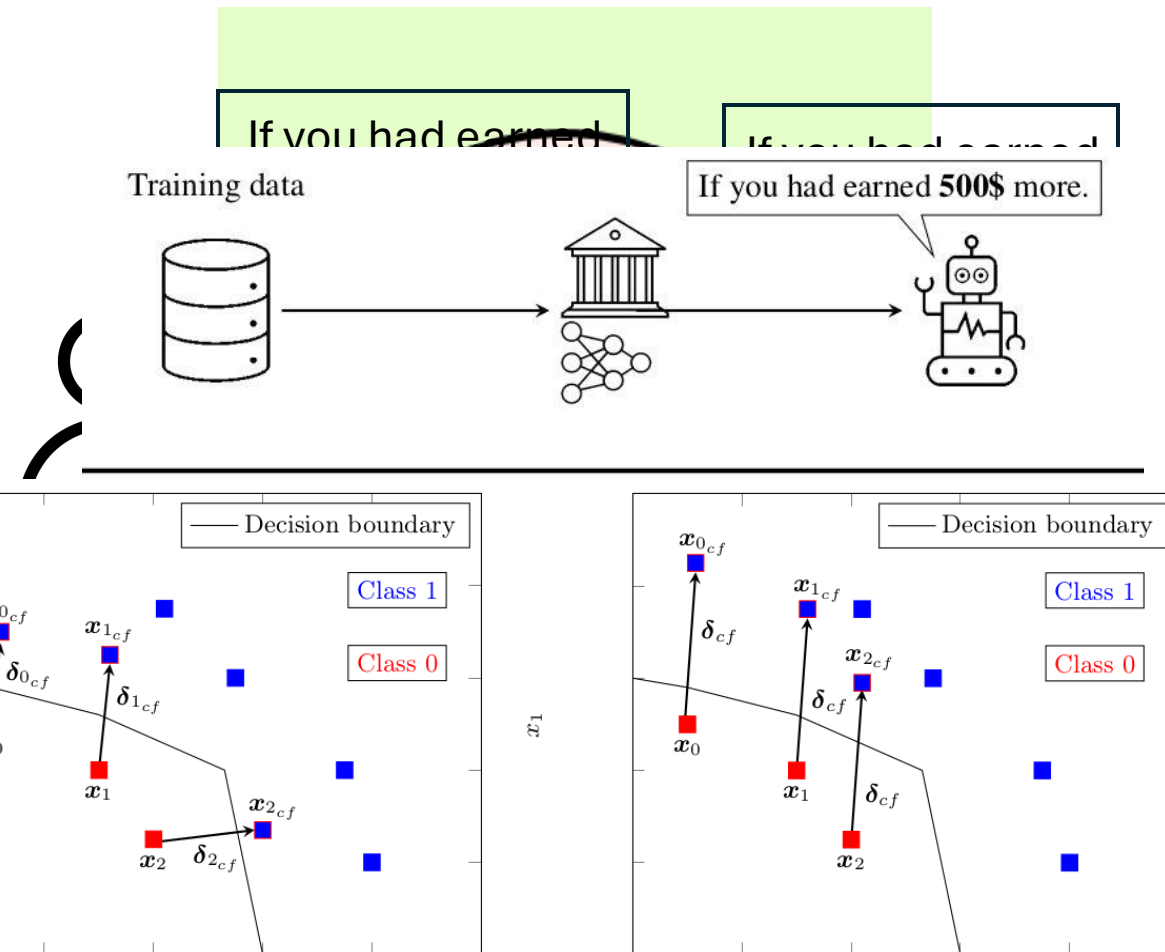
(b) Sub-cluster with $d(\vec{\delta}_{cf_i}, \vec{\delta}_{cf_j}) = \|\theta(\vec{\delta}_{cf_i}) - \theta(\vec{\delta}_{cf_j})\|_2$

Open Questions

- Fairness?
- Robustness?
 - => Outliers, poisonous instances, etc.
- Computational complexity and limitations?
 - => Formal analysis
- Python framework/implementation
- ...

Summary

- **Robustness**
- **Fairness**
- CFs can be **manipulated!**
 - How to defend?
- CFs for groups of instances -- i.e. **group CFs**
 - Gaining popularity
- Many things not covered in this tutorial!



Questions and Discussions

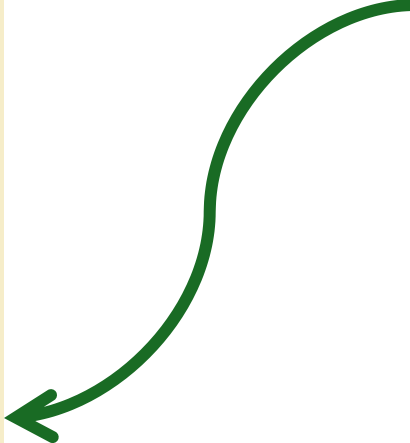
Ask us anything!

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Hands-On: Counterfactuals in Python



<https://tinyurl.com/ycx3scaj>